

Assessment of Drinking-Water Quality, Climate Variability and Water-Related Disease Patterns in Akure Metropolis, Ondo State, Nigeria

Adenike Abigail Orebanjo^{1*}, Iyabo Florence Ogunmakinwa², Funmini Joy Adebayo-Salami¹

¹Department of Public Health, College of Public Health, Wesley University Ondo, Ondo State.

²Department of Community Health, College of Community Health, Wesley University Ondo, Ondo State.

*Corresponding author email address: adenikeorebanjo628@gmail.com

Abstract	Article History
Access to safe drinking water is essential for public health, particularly in rapidly growing urban communities where environmental and microbiological contamination may increase the risk of water-related diseases. This study assessed drinking-water quality, climatic variability and selected water-related disease patterns in Akure Metropolis, Ondo State, Nigeria, using secondary data from available water-quality, climate and disease-surveillance sources. Physicochemical and microbiological characteristics of drinking-water samples were assessed, while rainfall, temperature and reported cases of diarrhoea and typhoid were examined descriptively. The water-quality findings showed that pH ranged from 4.50 to 7.93, turbidity from 0.11 to 6.60 NTU and nitrate from 2.10 to 61.00 mg/L. <i>Escherichia coli</i> ranged from 0 to 57 CFU/100 mL, while total coliforms ranged from 0 to 99 CFU/100 mL. Eleven of 14 pH observations (78.6%) fell outside the operational screening range of 6.5–8.5, while four of 11 <i>E. coli</i> observations (36.4%) exceeded the non-detectable drinking-water criterion; one nitrate observation and one turbidity observation also exceeded their respective reference criteria. Climatic data showed considerable variation in rainfall and temperature during the study period. Disease-surveillance records showed reported cases of diarrhoea and typhoid in Akure North and Akure South during June and July 2023. Overall, the findings indicate variability in drinking-water quality and the presence of microbiological contamination in some water sources, highlighting the need for regular water-quality monitoring, improved protection of water sources, appropriate treatment of contaminated water and strengthened environmental-health surveillance in Akure Metropolis. Keywords: Drinking water; water quality; microbiological contamination; climate variability; diarrhoea; typhoid; disease surveillance; public health; Akure Metropolis; Nigeria.	Received: 30 Jul 2026 Accepted: 31 Aug 2026 Published: 12 Sept 2026  Scan QR code to view* License: CC BY 4.0  Open Access article
How to cite this paper: Orebanjo, A. A., Ogunmakinwa, I. F., & Adebayo-Salami, F. J. (2026). Assessment of drinking-water quality, climate variability and water-related disease patterns in Akure Metropolis, Ondo State, Nigeria. <i>Journal of Pollution Monitoring, Evaluation Studies and Control</i>, 5(2), 276–290. https://doi.org/10.54117/8rk72190	

1. Introduction

Safe and reliable drinking water is fundamental to human health, social development and economic well-being. Water is required for drinking, food preparation, hygiene and sanitation, yet the presence of physical, chemical and microbiological hazards can turn an essential resource into an important route of human exposure to disease. The World Health Organization (WHO) emphasizes that drinking-water safety should be managed from the source through treatment and distribution to the point of consumption, with health protection based on risk assessment, preventive management and surveillance (World Health Organization [WHO], 2022). More recently, WHO has further emphasized the importance of water-safety planning, sanitary inspection, microbial monitoring and independent surveillance, particularly for small and decentralized water supplies (WHO, 2024, 2026). Globally, progress in access to drinking water has improved, but important inequalities remain, particularly in low- and middle-income settings where population growth, urbanization, inadequate infrastructure and environmental

contamination continue to affect the reliability and safety of water supplies (WHO & UNICEF, 2023; WHO, 2022).

Nigeria faces many of these challenges. Rapid population growth and urban expansion have increased demand for water while the availability and reliability of centralized water supply remain inadequate in many communities. Consequently, households frequently depend on groundwater sources, including boreholes and hand-dug wells, as well as commercially packaged water. National evidence shows substantial variation in drinking-water quality across different sources and geographical settings, with contamination associated with sewage, waste disposal, hydrogeological processes, agricultural activities and other anthropogenic pressures (Ighalo & Adeniyi, 2020). A recent systematic review of groundwater-quality research in Nigeria similarly identified continuing challenges in monitoring, data availability and the protection of groundwater resources as industrialization and population growth place increasing pressure on water systems (Omeka *et al.*, 2024). At the household level,

the type and quality of drinking-water source also have implications for health. Using nationally representative Nigerian household data, Adamu *et al.* (2022) reported that households relying on unimproved water sources had greater odds of reporting diarrhoea than households using improved sources. These findings demonstrate that access to water should not be considered only in terms of availability; the quality and safety of that water are equally important.

The quality of groundwater is influenced by both natural processes and human activities. Geological characteristics, aquifer conditions, land use, waste disposal, sanitation practices and proximity to potential contamination sources can influence the physicochemical and microbiological properties of groundwater. Studies from different parts of Nigeria have documented variation in pH, electrical conductivity, total dissolved solids, turbidity, nitrate and other chemical characteristics of groundwater and domestic water sources (Egbueri *et al.*, 2020; Gbadebo, 2021; Adesakin *et al.*, 2020). In some locations, groundwater has been found to be acidic or to contain elevated concentrations of particular chemical constituents, while microbiological contamination has also been documented in groundwater sources (Afonne *et al.*, 2020; Ganiyu *et al.*, 2021). Such findings are important because water that appears clear and acceptable based on physical characteristics may still contain microorganisms capable of causing gastrointestinal and other infections.

Microbiological quality is particularly important in assessing drinking-water safety. *Escherichia coli* and other faecal indicator organisms are widely used to indicate contamination associated with faecal material and the possible presence of enteric pathogens. Research in Nigeria has repeatedly demonstrated the occurrence of coliform organisms and other bacteria in household and packaged drinking-water sources (Adesakin *et al.*, 2020; Udoh *et al.*, 2021). A systematic review and meta-analysis of packaged drinking water in Nigeria found considerable contamination with total coliforms and faecal indicator bacteria, including *E. coli*, and identified treatment failure and inadequate regulatory oversight as important concerns (Udoh *et al.*, 2021). Other Nigerian studies have also identified *E. coli*, *Klebsiella*, *Enterobacter* and other organisms in drinking-water sources, demonstrating that microbiological contamination is not limited to one type of water supply (Ganiyu *et al.*, 2021; Rabiou *et al.*, 2022). In southwestern Nigeria, contamination has also been documented in water systems serving urban populations, emphasizing the continuing importance of microbiological surveillance (Oyewale *et al.*, 2024).

Packaged water is an important part of the Nigerian drinking-water landscape. Sachet water is widely consumed because it is relatively affordable, portable and readily available, particularly in urban environments. However, its widespread availability does not necessarily guarantee consistent microbiological or chemical quality. A systematic review of sachet-water studies across Nigeria identified substantial variation in physicochemical and microbiological quality and reported the occurrence of organisms including *E. coli*, *Salmonella*, *Enterobacter*, *Staphylococcus* and *Enterococcus* species (Agbasi *et al.*, 2024). Udoh *et al.* (2021) similarly found evidence of microbial contamination in packaged drinking water and highlighted the need for stronger monitoring and regulatory oversight. Individual investigations have also reported microbiological or chemical concerns in sachet water in different Nigerian communities (Kusa & Joshua, 2023; Alabi *et al.*, 2024). These findings do not mean that all packaged water is

unsafe; rather, they demonstrate the importance of evaluating water quality using measurable microbiological and physicochemical indicators rather than relying solely on the appearance, packaging or presumed source of the water.

Evidence from southwestern Nigeria provides further justification for examining drinking-water quality at the local level. Akure and its surrounding communities depend on a mixture of groundwater and commercially packaged water, and previous investigations have documented variability in both the physicochemical and microbiological characteristics of local water resources. An earlier investigation of drinking-water sources in Akure reported microbiological contamination among several water sources (Uzuegbu *et al.*, 1996), while more recent groundwater research in the metropolis has assessed multiple physicochemical characteristics and demonstrated spatial variation in groundwater quality (Afolabi *et al.*, 2022). Another investigation of groundwater sources in Akure identified enteric bacteria in both well and borehole water and reported potential microbial health risks associated with consumption (Olalemi *et al.*, 2021). These studies provide important local evidence, but their individual sampling designs, parameters and periods of observation differ. Consequently, bringing together available water-quality observations can provide a broader description of the environmental conditions affecting drinking-water safety in the metropolis.

Water quality is also affected by environmental and climatic conditions. Rainfall can influence the movement of contaminants from surrounding land into water bodies and shallow groundwater, while heavy rainfall and flooding can overwhelm sanitation systems and increase opportunities for faecal contamination. Conversely, prolonged dry periods may influence pathogen concentrations and water availability. The relationship between rainfall and diarrhoeal disease is therefore not necessarily linear and may depend on the amount and timing of rainfall, antecedent environmental conditions, water infrastructure and sanitation practices. A systematic review and meta-analysis by Kraay *et al.* (2020) found evidence supporting a concentration–dilution relationship in which rainfall following dry periods may flush pathogens into water systems, whereas rainfall following wetter periods may dilute pathogen concentrations. Seasonal changes in water contamination have also been documented in Nigerian urban settings. In Port Harcourt, for example, thermotolerant-coliform contamination varied between dry and rainy seasons across different drinking-water sources (Kumpel *et al.*, 2017).

Climate variability is particularly relevant to public health in rapidly expanding urban environments. Changes in rainfall patterns, flooding, drought and temperature can affect water availability, water treatment systems, sanitation infrastructure and the movement of pathogens through the environment (WHO, 2022). The relationship between environmental conditions and infectious disease is consequently influenced by several interacting factors, including water source, sanitation, hygiene, human behaviour and the resilience of local infrastructure. This makes local environmental monitoring important because broad regional climate patterns may not adequately describe the conditions experienced by individual communities.

Water-related diseases remain an important public-health concern in Nigeria. Diarrhoeal disease can result from exposure to contaminated water, food and environments, and inadequate sanitation and hygiene can facilitate transmission. In a

community-based study in southeastern Nigeria, Nwokoro *et al.* (2020) identified sanitation and hygiene practices among factors associated with diarrhoeal morbidity. At the national level, Adamu *et al.* (2022) demonstrated an association between drinking-water source and diarrhoea among Nigerian households. Evidence from intervention studies also shows that improvements in drinking-water quality, sanitation and hygiene can reduce diarrhoeal disease risk (Wolf *et al.*, 2022). These findings reinforce the importance of considering drinking-water quality within a broader environmental-health framework rather than treating water quality as an isolated laboratory issue.

Other enteric infections, including typhoid fever, are also closely linked to environmental conditions and the quality of water and sanitation systems. The transmission of *Salmonella enterica* serovar Typhi is associated with faecal contamination and inadequate sanitation, making safe water, sanitation and hygiene important components of prevention. More broadly, studies of enteric microorganisms in Nigerian water sources continue to demonstrate the potential for contaminated water to contribute to gastrointestinal infection risks (Olalemi *et al.*, 2021; Rabiou *et al.*, 2022). Viral pathogens also warrant consideration. For example, Ekundayo *et al.* (2021) reviewed human norovirus contamination of water sources and highlighted the importance of contaminated water in the environmental transmission of enteric viruses. More recent work on the epidemiology of norovirus in Nigeria further demonstrates the continuing relevance of enteric infections to public health surveillance (Chigor *et al.*, 2024).

The public-health significance of drinking-water quality therefore extends beyond whether individual physicochemical parameters fall within acceptable ranges. Microbial contamination, chemical contamination and environmental conditions can interact to influence exposure and health risk. For example, water may have acceptable pH, conductivity or total dissolved solids while still containing faecal indicator organisms. Conversely, elevated turbidity may indicate inadequate treatment or conditions that interfere with effective disinfection. Studies from Nigeria have documented this distinction between physicochemical acceptability and microbiological safety (Adesakin *et al.*, 2020; Udoh *et al.*, 2021). In addition, emerging contaminants have received increasing attention. Recent research in southwestern Nigeria has identified mycotoxins in different drinking-water sources, including groundwater and sachet water, illustrating the expanding range of contaminants that may require consideration in future water-safety programmes (Koko *et al.*, 2024). Other emerging contaminants, such as per- and polyfluoroalkyl substances, have also become an increasing concern in African drinking-water research (Adewuyi & Li, 2024).

Akure provides an important setting in which to examine these issues because of its growing urban population, dependence on multiple drinking-water sources and changing environmental conditions. Existing studies have generated valuable information on groundwater chemistry, microbiological contamination and packaged-water quality, but the evidence is distributed across different studies, sampling periods and water-source categories. This makes it difficult to obtain a broader public-health perspective from any single study. In addition, environmental and disease-surveillance information are often collected through separate systems, limiting opportunities to examine drinking-water quality alongside climatic and disease patterns. A more integrated assessment can therefore help identify recurring water-quality concerns, describe environmental variability and place

available disease-surveillance observations within their local context.

The present study was designed to address this need by integrating available secondary data on drinking-water quality, climate and selected water-related disease surveillance in Akure Metropolis, Ondo State, Nigeria. The study evaluates physicochemical and microbiological characteristics of available drinking-water observations, compares selected parameters across water sources where the underlying data permit, assesses observations against relevant drinking-water criteria, describes rainfall and temperature patterns, and examines available diarrhoea and typhoid surveillance records alongside environmental conditions. By bringing these strands of evidence together, the study seeks to provide a clearer public-health perspective on drinking-water safety and environmental conditions in Akure Metropolis and to generate evidence that can support water-quality monitoring, environmental-health planning and future research in the area.

2. Materials and Methods

2.1 Study design

A retrospective descriptive study design was used to integrate secondary data on drinking-water quality, climatic conditions and selected disease-surveillance indicators in Akure Metropolis, Ondo State, Nigeria. The study combined physicochemical and microbiological water-quality observations with meteorological and epidemiological data to provide an integrated assessment of environmental conditions relevant to public health.

2.2 Study Area

The study was conducted in Akure Metropolis, the capital of Ondo State in southwestern Nigeria. The metropolitan area comprises Akure South and Akure North Local Government Areas. Akure has a tropical climate characterised by distinct wet and dry seasons, with rainfall occurring predominantly between April and October. The area is underlain mainly by Precambrian Basement Complex rocks, including migmatite, granite and charnockite, which influence groundwater occurrence and chemistry.

2.3 Data Sources

Three categories of secondary data were used:

Water-quality data: Physicochemical and microbiological observations were compiled from available published drinking-water and groundwater investigations in Akure Metropolis (Table1). The coded 2017 matched dataset was traced to Daramola and Akindureni (2017), which reported monthly results for April, June and August 2017 from BORBDA borehole water, TISCO borehole water and ODSWC (Ukere) tap water. For clarity, BORBDA denotes the Benin-Owena River Basin Development Authority; TISCO denotes TISCO Pharmaceutical Company; and ODSWC_Ukere denotes Ondo State Water Corporation (Ukere tap-water supply). The variables extracted across the compiled water-quality dataset included pH, temperature, turbidity, electrical conductivity (EC), total dissolved solids (TDS), nitrate, nitrite, chloride, calcium, magnesium, hardness, iron, sulphate, fluoride, chromium, potassium, *Escherichia coli*, total coliforms and faecal streptococci.

Climate data: Monthly meteorological observations for Akure were incorporated into the study database. The variables included rainfall, mean temperature, minimum temperature, maximum temperature, rainy days, relative humidity and wind speed where available. The climate dataset covered January 2021 to June 2023.

Disease-surveillance data: Epidemiological surveillance reports. Reported diarrhoea and typhoid cases for Akure North information was obtained from Ondo State disease-surveillance and Akure South were extracted for analysis.

Table 1. Secondary Data Sources and Integration Structure

Dataset component	Documented source	Coverage	Role in analysis
Water quality — matched 2017 subset	Daramola & Akindureni (2017)	April, June and August 2017; BORBDA, TISCO and ODSWC (Ukere)	Three-period matched observations used for the Friedman test and the corresponding source-group descriptive results.
Water quality — compiled observations	Available published Akure drinking-water/groundwater observations	Heterogeneous periods and source categories; parameter availability varied	Descriptive compilation. Record-level source attribution was retained where available; missing parameter values were not imputed.
Climate	Akure monthly meteorological observations	January 2021–June 2023	Descriptive summary of rainfall, temperature and other available climate variables. The present manuscript does not identify the provider/station by name.
Disease surveillance	Ondo State disease-surveillance reports	June–July 2023; Akure North and Akure South	Descriptive summary of reported diarrhoea and typhoid cases.

2.4 Data Extraction and Management

Relevant observations were extracted into a structured master database. For each water-quality observation, the water source, location, parameter, numerical value, measurement unit, sampling period and original source were recorded where available. The datasets were combined by aligning parameter names, units, water-source categories and observation periods while retaining the original source and sampling-period information. Measurements were harmonised into consistent units before analysis. Because the underlying studies did not report every parameter for every observation, the analytical sample size (N) differs by parameter; N represents the number of non-missing observations available for that parameter after harmonisation. Missing observations were retained as missing values and were not replaced with zero or estimated values.

2.5 Water-Quality Assessment

Water-quality observations were assessed using physicochemical and microbiological indicators. Descriptive statistics were calculated for the available observations, including mean, standard deviation, median, minimum and maximum values. The results were compared with applicable drinking-water guideline criteria to identify observations exceeding reference values.

2.6 Climate and Disease Data Analysis

Monthly rainfall and temperature data were summarised to describe climatic patterns during the study period. Disease-

surveillance data were summarised according to disease, month and Local Government Area. Climate and disease observations were examined descriptively to characterise their temporal occurrence within the study period.

2.7 Statistical Analysis

Statistical analysis was performed using descriptive and non-parametric methods appropriate to the structure of the datasets. For the matched 2017 water-quality subset, the three sampling months (April, June and August) were treated as repeated blocks and the three water-source groups (BORBDA, TISCO and ODSWC_Ukere) as the related groups for each parameter. The Friedman test was applied to each parameter with complete matched observations, and Kendall's coefficient of concordance (W) was used to describe the strength of agreement in the rankings. Statistical significance was set at $p < .05$. The inferential analysis was restricted to this matched three-period dataset and was not applied to the heterogeneous compiled observations.

3. Results

3.1 Climate Characteristics

The climate dataset comprised monthly observations for Akure covering January 2021 to June 2023. Rainfall and temperature characteristics are summarised in Table 2.

Table 2. Monthly and Annual Climate Characteristics in Akure, 2021–2023

Year	Months available	Total rainfall (mm)	Mean monthly temperature (°C)	Minimum temperature (°C)	Maximum temperature (°C)	Rainy days
2021	12	2,459.5	25.93	17.6	38.0	293
2022	12	1,325.3	25.48	17.0	40.6	282
2023	6	393.4	26.73	17.0	40.1	103

The highest annual rainfall was recorded in 2021 (2,459.5 mm), while 2022 recorded 1,325.3 mm (Fig. 1). A total of 393.4 mm was recorded during the available 2023 period. Mean monthly temperature ranged from 25.48°C in 2022 to 26.73°C during the available 2023 period. Rainy days were highest in 2021 (293 days).

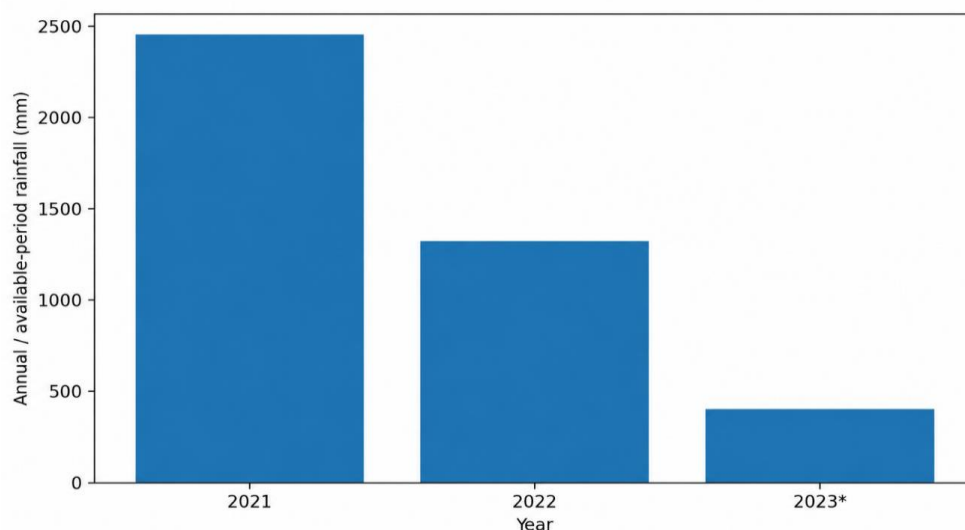


Figure 1. Annual Rainfall and Mean Temperature in Akure, 2021–2023.

The figure demonstrates marked variation in annual rainfall across the study period, with the highest total rainfall recorded in 2021 (2,459.5 mm) and a lower total in 2022 (1,325.3 mm). The available 2023 observations recorded 393.4 mm, although only six months were available for that year. Mean temperature remained relatively stable across the period, ranging from 25.48°C in 2022 to 26.73°C in the available 2023 observations. Overall, the figure demonstrates considerable rainfall variability while temperature remained comparatively consistent.

3.2 Physicochemical and Microbiological Characteristics of Water

The compiled water-quality observations were assessed using physicochemical and microbiological indicators. The overall descriptive characteristics are presented in Table 3.

The observed pH ranged from 4.50 to 7.93, while turbidity ranged from 0.11 to 6.60 NTU. Nitrate concentrations ranged from 2.10 to 61.00 mg/L. Microbiological indicators showed substantial variation, with *E. coli* ranging from 0 to 57 CFU/100 mL and total coliforms from 0 to 99 CFU/100 mL (Fig. 2)

Table 3. Overall Physicochemical and Microbiological Characteristics of Water-Quality Observations

Parameter	Unit	N	Mean $\pm$ SD	Median	Minimum	Maximum
pH	—	14	6.13 $\pm$ 0.84	6.05	4.50	7.93
Temperature	°C	14	28.65 $\pm$ 1.53	28.73	26.00	30.90
Turbidity	NTU	14	1.51 $\pm$ 1.75	1.08	0.11	6.60
Electrical conductivity	μ S/cm	11	258.18 $\pm$ 134.13	179.00	133.00	479.66
Total dissolved solids	mg/L	12	169.80 $\pm$ 100.54	118.59	18.00	342.00
Nitrate	mg/L	14	14.75 $\pm$ 18.19	4.65	2.10	61.00
<i>E. coli</i>	CFU/100 mL	11	6.82 $\pm$ 16.95	0.00	0.00	57.00
Total coliform	CFU/100 mL	11	18.18 $\pm$ 28.85	10.00	0.00	99.00
Faecal streptococci	CFU/100 mL	11	4.18 $\pm$ 11.14	0.00	0.00	37.00

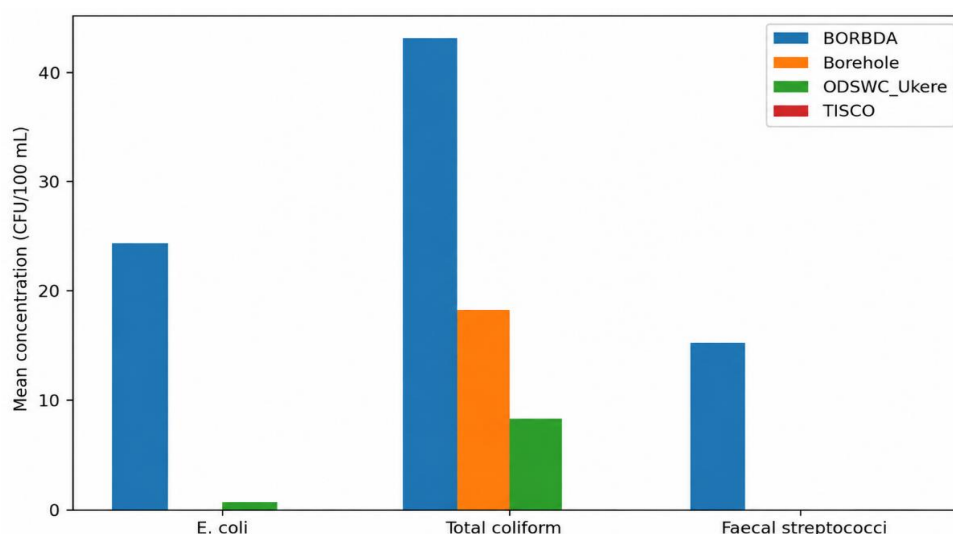


Figure 2. Microbiological characteristics of water sources in Akure Metropolis.

The microbiological profile demonstrates substantial differences among the water sources. BORBDA observations recorded the highest mean *E. coli*, total coliform and faecal-streptococci concentrations, whereas *E. coli* was not detected in the borehole, hand-dug-well and TISCO observations represented in the table. The presence of faecal indicator organisms, particularly *E. coli*, is important because it indicates potential microbiological contamination of drinking-

water sources. These results support the need for continued microbiological monitoring of drinking-water sources in the study area.

3.3 Water-Quality Characteristics by Water Source

The major physicochemical and microbiological characteristics according to water source are presented in Table 4.

Table 4. Key Water-Quality Characteristics by Water Source

Parameter	BORBDA	Borehole	Hand-dug well	ODSWC_Ukere	TISCO
pH	6.07 ± 0.21	6.51 ± 0.73	7.93	5.17 ± 0.59	6.03 ± 0.21
Temperature (°C)	28.10 ± 2.52	27.58 ± 0.91	28.73	30.00 ± 1.04	29.27 ± 0.68
Turbidity (NTU)	0.45 ± 0.37	3.25 ± 2.43	2.48	0.78 ± 0.50	0.64 ± 0.70
EC (µS/cm)	143.33 ± 11.06	186.33	479.66	405.67 ± 9.29	175.67 ± 3.06
TDS (mg/L)	96.03 ± 7.41	193.67 ± 163.72	—	271.80 ± 6.23	117.70 ± 2.05
Nitrate (mg/L)	2.80 ± 0.75	35.14 ± 22.38	3.67	15.03 ± 9.05	2.93 ± 0.78
<i>E. coli</i> (CFU/100 mL)	24.33 ± 28.31	0.00	0.00	0.67 ± 1.15	0.00
Total coliform (CFU/100 mL)	43.67 ± 48.29	18.00	26.00	8.33 ± 13.58	0.00
Faecal streptococci (CFU/100 mL)	15.33 19.09	0.00	0.00	0.00	0.00

The water sources showed differences in several physicochemical and microbiological characteristics. BORBDA observations had the highest mean *E. coli*, total coliform and faecal-streptococci values, whereas no *E. coli*

was observed in the borehole, hand-dug well and TISCO observations represented in this table. Electrical conductivity and TDS were comparatively higher in ODSWC_Ukere and the hand-dug well observations (Fig. 3).

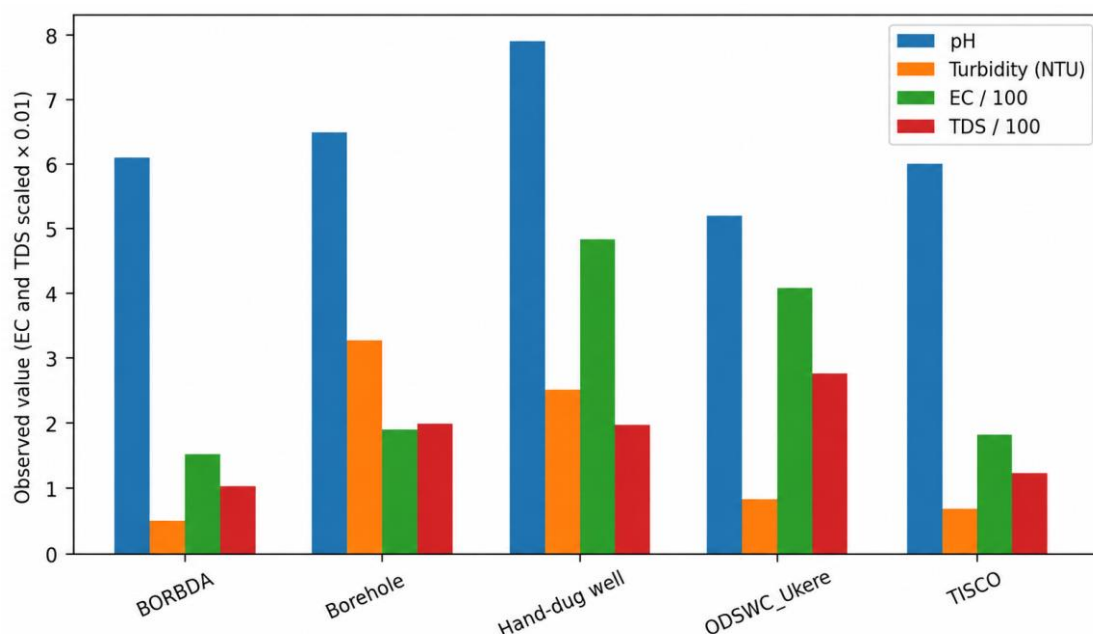


Figure 3. Key physicochemical characteristics of water sources in Akure Metropolis.

The figure shows differences in the physicochemical characteristics of water among the evaluated sources. Electrical conductivity and TDS were particularly elevated in the ODSWC_Ukere observations, while the hand-dug well showed the highest single EC value. Turbidity was comparatively higher in borehole and hand-dug-well observations than in BORBDA, ODSWC_Ukere and TISCO

observations. The variation suggests differences in mineral content, dissolved substances and physical characteristics among the water sources.

3.4 Inferential Analysis of Water-Quality Parameters

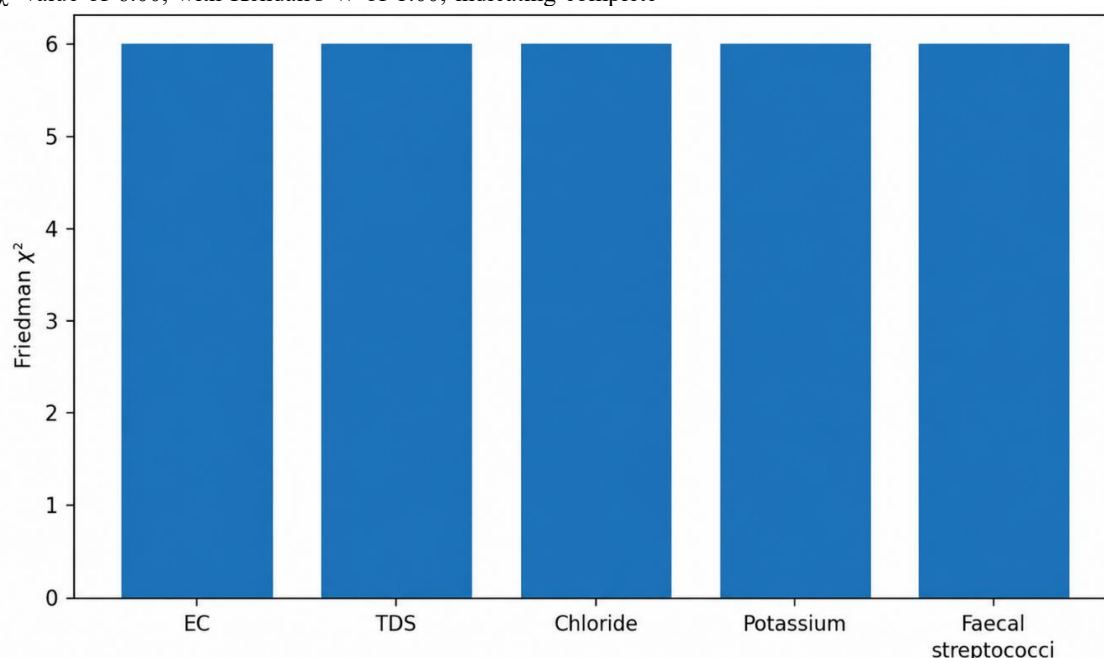
The Friedman test was applied to the matched 2017 water-quality dataset. The results are presented in Table 5.

Table 5. Friedman Test of Water-Quality Parameters in the Matched 2017 Dataset

Parameter	Friedman χ^2	p-value	Kendall's W	Result
pH	4.667	.097	0.778	Not significant
Temperature	0.667	.717	0.111	Not significant
EC	6.000	.050	1.000	Significant
Turbidity	0.667	.717	0.111	Not significant
TDS	6.000	.050	1.000	Significant
Hardness	4.667	.097	0.778	Not significant
Calcium	4.667	.097	0.778	Not significant
Magnesium	4.667	.097	0.778	Not significant
Sulphate	2.364	.307	0.394	Not significant
Chromium	2.667	.264	0.444	Not significant
Fluoride	3.455	.178	0.576	Not significant
Chloride	6.000	.050	1.000	Significant
Nitrate	4.667	.097	0.778	Not significant
Potassium	6.000	.050	1.000	Significant
<i>E. coli</i>	5.600	.061	0.933	Not significant
Total coliform	3.818	.148	0.636	Not significant
Faecal streptococci	6.000	.050	1.000	Significant

The figure presents the five water-quality parameters for which the Friedman test reached the prespecified significance level of $p = .050$: electrical conductivity, TDS, chloride, potassium and faecal streptococci (Fig. 4). Each parameter produced a Friedman χ^2 value of 6.00, with Kendall's W of 1.00, indicating complete

concordance in the ranking pattern across the matched observations. However, these results should be interpreted cautiously because the inferential analysis was based on only three matched sampling periods.

**Figure 4.** Water-quality parameters showing significant Friedman-test results.

3.5 Water-Quality Guideline Assessment

The observed water-quality parameters were compared with applicable drinking-water reference criteria. The proportions outside the specified criteria are presented in Table 6.

The highest proportion outside the specified range (Fig. 5) was observed for pH (78.6%), followed by *E. coli* (36.4%). One nitrate observation and one turbidity observation exceeded their respective reference criteria.

Table 6. Water-Quality Observations outside Reference Criteria

Parameter	Reference criterion	N assessed	Outside criterion	Percentage (%)
<i>E. coli</i>	Not detectable	11	4	36.4
Nitrate	50 mg/L	14	1	7.1
Nitrite	3 mg/L	3	0	0.0
Turbidity	5 NTU	14	1	7.1
pH	6.5–8.5	14	11	78.6

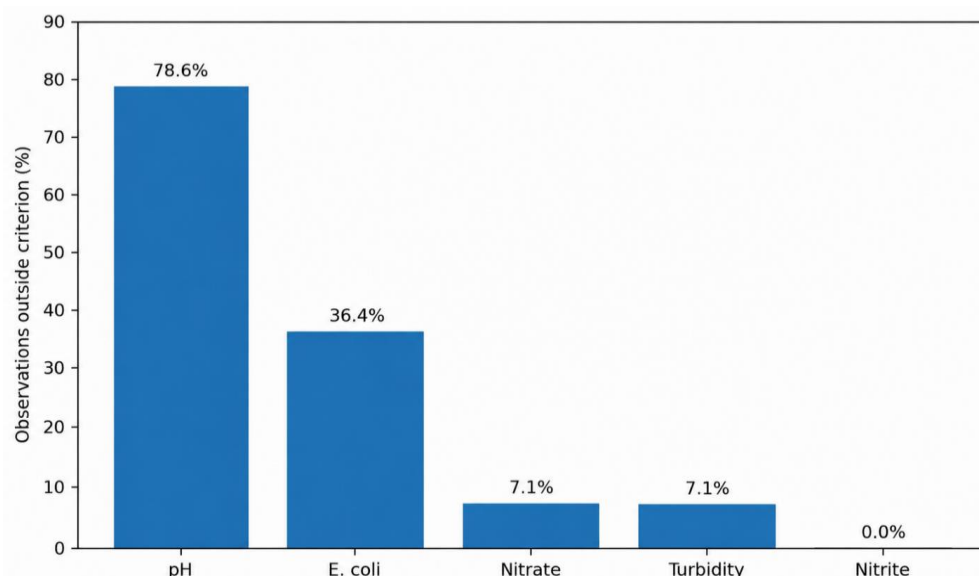


Figure 5. Proportion of Water-Quality Observations outside Reference Criteria

The greatest proportion of observations outside the operational reference range was observed for pH (78.6%), followed by *E. coli* (36.4%). Nitrate and turbidity each had 7.1% of observations outside their specified criteria, while no nitrite observation exceeded the criterion used in the analysis. The pH finding should be interpreted as an operational screening observation rather than a WHO health-based exceedance, while the *E. coli* finding represents the more direct microbiological public-health concern.

3.6 Disease-Surveillance Findings

Disease-surveillance observations for Akure North and Akure South during June and July 2023 are presented in Table 7.

Reported diarrhoea and typhoid cases varied between the two LGAs and across the two months (Fig. 6). In June, Akure South recorded 59 diarrhoea cases and 202 typhoid cases, while Akure North recorded six typhoid cases. In July, Akure North recorded 26 diarrhoea cases and 152 typhoid cases, while Akure South recorded 37 diarrhoea cases.

Table 7. Reported Diarrhoea and Typhoid Cases in Akure North and Akure South, June–July 2023

Month	LGA	Diarrhoea cases	Typhoid cases
June 2023	Akure North	—	6
June 2023	Akure South	59	202
July 2023	Akure North	26	152
July 2023	Akure South	37	—

Note. — indicates that no case count was reported or available in the source surveillance record; it does not indicate zero cases.

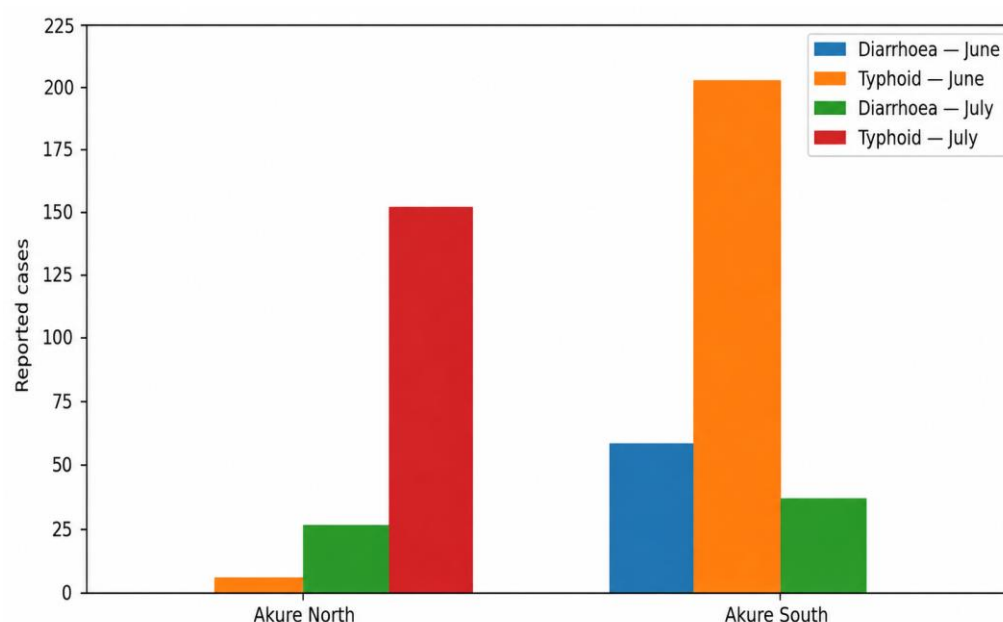


Figure 6. Reported diarrhoea and typhoid cases in Akure North and Akure South, June–July 2023.

The figure demonstrates substantial differences in reported disease counts between Akure North and Akure South during June and July 2023. In June, Akure South recorded 59 diarrhoea cases and 202 typhoid cases, while Akure North recorded six typhoid cases and no diarrhoea count was available in the record. In July, Akure North recorded 26 diarrhoea cases and 152 typhoid cases, while Akure South recorded 37 diarrhoea cases and no typhoid count was available in the record. The differences should be interpreted as patterns in reported surveillance records rather than evidence of causal differences between the LGAs.

3.7 Climate and Disease-Surveillance Characteristics

The climate and disease-surveillance observations available for June 2023 are presented together in Table 8.

Table 8. Climate and Disease-Surveillance Characteristics for June 2023

Variable	June 2023
Rainfall	40.5 mm
Mean temperature	25.08°C
Rainy days	6
Diarrhoea cases in Akure	59
Typhoid cases in Akure	208
Ondo State diarrhoea cases	291
Ondo State typhoid cases	717
Akure share of state diarrhoea cases	20.27%
Akure share of state typhoid cases	29.01%

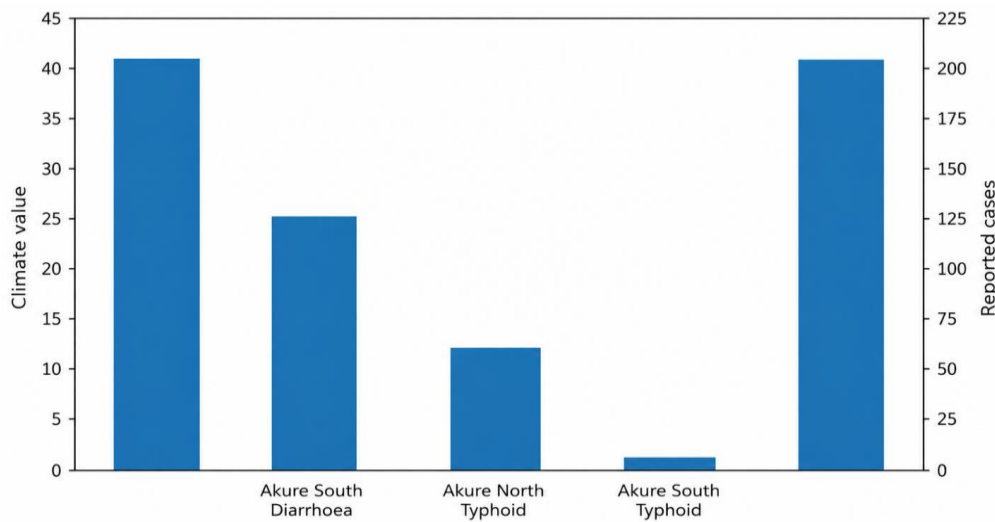


Figure 7. June 2023 Climate Conditions and Reported Disease Cases

4. Discussion

The present study provides an integrated assessment of drinking-water quality, climatic conditions and selected disease-surveillance observations in Akure Metropolis. The major finding was the presence of substantial variability in both physicochemical and microbiological water-quality characteristics, with microbiological contamination representing an important public-health concern. In particular, *Escherichia coli* was detected in some of the assessed water observations, while total coliforms and faecal streptococci also occurred in several samples. This finding is consistent with evidence from other parts of Nigeria showing that contamination of drinking-water sources remains an important environmental-health concern. Udoh *et al.* (2021), in a systematic review and meta-analysis of packaged drinking water in Nigeria, reported contamination with total coliforms and faecal indicator bacteria, including *E. coli*, and concluded that stronger monitoring of packaged drinking-water quality is required. Similarly, Agbasi *et al.* (2024) reviewed more

than 50 studies of sachet water across Nigeria and identified substantial regional variation in physicochemical and microbiological quality, including the occurrence of *E. coli*, *Salmonella*, *Enterobacter* and other organisms. These findings support the observation in the present study that the microbiological quality of drinking water cannot be assumed from the type or appearance of the water source alone.

During June 2023, rainfall was 40.5 mm, mean temperature was 25.08°C and six rainy days were recorded. The reported Akure cases accounted for 20.27% of the available Ondo State diarrhoea cases and 29.01% of typhoid cases for the same period. In June 2023, the available climate record showed 40.5 mm of rainfall, a mean temperature of 25.08°C and six rainy days (Fig. 7). During the same month, Akure South recorded 59 diarrhoea cases and 202 typhoid cases, while Akure North recorded six typhoid cases. The figure provides a descriptive comparison of the available environmental and surveillance observations. Because only one month provides a direct verified climate–disease match, the pattern should not be interpreted as a statistical association or causal relationship.

beverages consumed in Nigeria also identified microbiological hazards in borehole/well water and sachet water, demonstrating that contamination occurs across multiple commonly consumed water sources (Oduori *et al.*, 2022). More recently, Agbasi *et al.* (2024) similarly reported faecal contamination across different geographical zones of Nigeria. The comparatively high proportion observed in the present dataset therefore reinforces the importance of routine microbiological monitoring of drinking-water sources in Akure.

The occurrence of total coliforms and faecal streptococci further supports evidence of microbiological variability among the water sources. In the present study, BORBDA observations recorded comparatively higher mean concentrations of *E. coli*, total coliforms and faecal streptococci than several of the other source categories. Such differences may reflect variations in source protection, sanitation conditions, treatment processes, storage and handling. Previous Nigerian investigations have similarly demonstrated differences in microbial quality between groundwater and packaged-water sources. Oduori *et al.* (2022) found evidence of important bacterial hazards in borehole and well water as well as packaged water, while Udoh *et al.* (2021) identified treatment failure and post-treatment contamination as important contributors to microbial contamination of sachet water. These observations suggest that monitoring should extend beyond the point of production or extraction to include storage, distribution and household handling where relevant.

The physicochemical results also showed considerable variation. Electrical conductivity, total dissolved solids, nitrate and turbidity varied across the assessed water sources, indicating differences in mineral content and environmental conditions. The relatively higher electrical conductivity and TDS observed in some groundwater sources are consistent with the understanding that groundwater chemistry is influenced by interactions between water and the geological materials through which groundwater moves. At the same time, elevated nitrate concentrations may reflect anthropogenic influences such as sanitation systems, waste disposal and agricultural activities, particularly where groundwater sources are vulnerable to surface-derived contamination. Nigerian groundwater studies have similarly reported substantial spatial variation in physicochemical characteristics and have highlighted the influence of geology and human activities on groundwater quality (Egbueri *et al.*, 2020; Ganiyu *et al.*, 2021). The present findings therefore reinforce the importance of assessing both physicochemical and microbiological characteristics when evaluating drinking-water safety.

The guideline assessment provides an important public-health interpretation of these findings. Although most of the observations for several physicochemical parameters were within the applicable reference values, four of the 11 *E. coli* observations exceeded the requirement for non-detectable *E. coli* in drinking water. One nitrate observation and one turbidity observation also exceeded their respective reference criteria. This pattern illustrates an important distinction between general physicochemical acceptability and microbiological safety. Water may have relatively acceptable conductivity, TDS or temperature while still presenting a microbiological concern. This observation agrees with previous reviews of Nigerian drinking water, which have reported situations in which physicochemical characteristics may appear acceptable while microbial contamination remains present (Udoh *et al.*, 2021; Agbasi *et al.*, 2024). The finding therefore supports the continued use of microbiological indicators

alongside physicochemical measurements in routine drinking-water surveillance.

The pH findings require a more cautious interpretation. Although 11 of the 14 observations fell outside the 6.5–8.5 operational range used in the present analysis, this should not be interpreted as evidence that these waters necessarily posed a direct health hazard because pH is primarily an operational water-quality parameter rather than a health-based toxicological limit at these concentrations. The variation in pH may nevertheless be relevant to water treatment, corrosion and the behaviour of other chemical constituents. Nigerian groundwater investigations have reported variation in pH associated with geological conditions and local environmental influences, supporting the interpretation that differences in pH should be considered within the hydrogeological context of each water source (Egbueri *et al.*, 2020; Ganiyu *et al.*, 2021).

The inferential analysis of the matched water-quality dataset showed statistically significant differences in electrical conductivity, TDS, chloride, potassium and faecal streptococci. The pattern suggests that some water-quality characteristics varied systematically across the matched sampling observations. The strong concordance observed for these parameters is also consistent with the expected relationship between conductivity and dissolved ionic constituents, since electrical conductivity reflects the concentration and mobility of dissolved ions in water. Similar relationships between physicochemical parameters have been reported in groundwater-quality investigations in Nigeria, where conductivity, TDS and major ions have been used to characterize hydrochemical variation (Egbueri *et al.*, 2020). The microbiological finding involving faecal streptococci is particularly relevant because it complements the *E. coli* and total-coliform findings and provides additional evidence of differences in microbial water quality.

The climatic findings showed considerable variation in rainfall and temperature during the study period, with higher annual rainfall recorded in 2021 than in 2022 and higher mean temperature during the available 2023 observations. Rainfall is relevant to water quality because precipitation can influence runoff, groundwater recharge and the movement of contaminants from contaminated surfaces into water sources. However, the relationship between rainfall and waterborne disease is complex and may vary according to local sanitation, drainage, water infrastructure and seasonal conditions. Kraay *et al.* (2020), in a systematic review and meta-analysis, reported evidence that rainfall can influence diarrhoeal disease through mechanisms involving pathogen mobilisation and changes in environmental exposure. The climatic variability observed in Akure therefore provides an important environmental context for interpreting water-quality and disease-surveillance patterns.

The disease-surveillance findings showed reported diarrhoea and typhoid cases in Akure North and Akure South during June and July 2023. Akure South recorded 59 diarrhoea cases and 202 typhoid cases in June, while Akure North recorded six typhoid cases during the same month. In July, 26 diarrhoea and 152 typhoid cases were reported in Akure North, while Akure South recorded 37 diarrhoea cases. These observations are consistent with the broader public-health literature linking inadequate water, sanitation and hygiene conditions with enteric disease transmission. Adamu *et al.* (2022), using nationally representative Nigerian data, found an association between drinking-water source and diarrhoeal illness, while Wolf *et al.* (2022)

demonstrated through a systematic review that improvements in drinking-water, sanitation and hygiene can reduce diarrhoeal disease. The surveillance observations in the present study therefore reinforce the relevance of maintaining safe drinking-water systems as part of broader enteric-disease prevention.

The observed disease pattern should, however, be interpreted within the wider environmental context rather than attributing reported cases directly to a particular water source. Diarrhoea and typhoid transmission may be influenced by multiple factors, including food hygiene, sanitation, household water storage, personal hygiene, healthcare-seeking behaviour and environmental exposure. Consequently, the coexistence of reported disease cases and water-quality concerns provides an important public-health signal but does not by itself establish a causal relationship. This broader interpretation is consistent with evidence from Nigeria showing that water, sanitation and hygiene conditions interact in determining enteric-disease risk (Adamu *et al.*, 2022; Oduori *et al.*, 2022).

Taken together, the findings indicate that drinking-water quality in Akure Metropolis is variable, with microbiological contamination representing the most important immediate public-health concern identified in the available data. The findings are broadly consistent with the wider Nigerian literature, which continues to report microbial contamination across groundwater and packaged-water sources despite improvements in water access (Udoh *et al.*, 2021; Agbasi *et al.*, 2024). The combination of physicochemical variability, microbiological contamination, climatic variability and reported enteric diseases highlights the importance of integrated environmental-health surveillance. Strengthening routine testing of drinking-water sources, improving protection of groundwater sources, maintaining effective treatment and distribution practices, and linking environmental monitoring with disease surveillance could provide a stronger basis for preventing water-related health risks in Akure Metropolis.

5. Conclusion

This study demonstrates that drinking-water quality in Akure Metropolis varies across water sources, with microbiological contamination representing an important public-health concern. The detection of *Escherichia coli*, total coliforms and faecal streptococci in some water-quality observations indicates the need for continued attention to the safety of drinking-water sources. Variations were also observed in physicochemical parameters, particularly electrical conductivity, total dissolved solids, nitrate and turbidity, while the climatic data demonstrated variation in rainfall and temperature during the study period. The disease-surveillance findings further showed the occurrence of reported diarrhoea and typhoid cases within Akure North and Akure South. Overall, the findings highlight the importance of regular monitoring of drinking-water quality, protection of groundwater sources, effective water treatment and strengthened environmental-health surveillance in Akure Metropolis. An integrated approach linking water-quality monitoring, environmental information and disease surveillance is recommended to support evidence-based public-health planning and reduce potential water-related health risks in the metropolis.

6. Recommendations

Based on the findings of this study, the following recommendations are made:

1. Routine water-quality monitoring: Regular physicochemical and microbiological testing of drinking-water sources in Akure Metropolis should be strengthened, with particular attention to *E.*

coli, total coliforms, turbidity, nitrate and other key indicators of water safety.

2. Protection of water sources: Boreholes, hand-dug wells and other groundwater sources should be adequately protected from sewage, refuse dumps, drainage channels and other potential sources of contamination.

3. Effective treatment of contaminated water: Water sources found to contain microbiological contamination should undergo appropriate treatment and disinfection before consumption, with treated water monitored regularly to ensure continued safety.

4. Strengthening packaged-water quality control: Producers and regulatory authorities should maintain regular quality assessment of packaged drinking water to ensure that water meets applicable drinking-water standards throughout production, storage and distribution.

5. Integrated environmental-health surveillance: Water-quality, climatic and disease-surveillance information should be routinely coordinated among relevant environmental-health and public-health authorities to support early identification of potential water-related health risks.

6. Public health education: Communities should be encouraged to adopt safe water-storage, handling and household treatment practices and to avoid activities that can contaminate wells and other drinking-water sources.

7. Further research and monitoring: Periodic studies covering a wider range of locations, water sources and environmental-health indicators should be conducted to strengthen the evidence base for drinking-water management and public-health planning in Akure Metropolis.

7. Limitation of the Study

This study has some limitations that should be considered when interpreting the findings. The analysis was based on secondary data obtained from existing water-quality, climatic and disease-surveillance sources; consequently, the study was dependent on the variables, sampling periods and measurements available in the original datasets. Differences in sampling locations, water sources and periods of data collection may also introduce variation in the underlying observations. In addition, the study did not directly control the original field-sampling and laboratory procedures used to generate the water-quality measurements. The disease-surveillance component was based on reported cases and therefore reflects available surveillance records rather than independently confirmed cases. Finally, the observational nature of the integrated dataset means that the findings describe patterns and associations but should not be interpreted as establishing direct causal relationships between water quality, climatic conditions and reported disease occurrence. These limitations do not diminish the value of the findings but indicate the need for continued prospective monitoring and standardized data collection in future studies.

References

- Adamu, I., Andrade, F. C. D., & Singleton, C. R. (2022). Availability of drinking water source and the prevalence of diarrhea among Nigerian households. *The American Journal of Tropical Medicine and Hygiene*, 107(4), 893–897. <https://doi.org/10.4269/ajtmh.21-0901>
- Adesakin, T. A., Oyewale, A. T., Bayero, U., Mohammed, A. N., Aduwo, I. A., Ahmed, P. Z., Abubakar, N. D., & Barje, I. B. (2020). Assessment of bacteriological quality and physico-chemical parameters of domestic water sources in Samaru community, Zaria, Northwest Nigeria. *Heliyon*, 6(8), e04773. <https://doi.org/10.1016/j.heliyon.2020.e04773>
- Adewuyi, A., & Li, Q. (2024). Per- and polyfluoroalkyl substances contamination of drinking water sources in Africa: Pollution

- sources and possible treatment methods. *Chemosphere*, 365, 143368. <https://doi.org/10.1016/j.chemosphere.2024.143368>
- Afolabi, O. O., Ogundana, A. K., & Ogundipe, I. E. (2022). Assessment and characterisation of groundwater quality in Akure metropolis, southwestern Nigeria. *Applied Water Science*, 12, Article 84. <https://doi.org/10.1007/s13201-021-01544-9>
- Afonne, O. J., Chukwuka, J. U., & Ifediba, E. C. (2020). Evaluation of drinking water quality using heavy metal pollution indexing models in an agrarian, non-industrialised area of South-East Nigeria. *Journal of Environmental Science and Health, Part A*, 55(12), 1406–1414. <https://doi.org/10.1080/10934529.2020.1796402>
- Agbasi, J. C., Ezugwu, A. L., Omeke, M. E., Ucheana, I. A., Aralu, C. C., Abugu, H. O., & Egbueri, J. C. (2024). More about making profits or providing safe drinking water? A state-of-the-art review on sachet water contamination in Nigeria. *Journal of Environmental Science and Health, Part C*, 42(4), 255–297. <https://doi.org/10.1080/26896583.2024.2319009>
- Alabi, O. A., Lawrence, O. E., Ayeni, F. E., & Olumurewa, J. A. V. (2024). Health risk and germ cell toxicity of five commercially available sachet waters in Nigeria: A public health concern. *Toxicology Research*, 13(4), tfae101. <https://doi.org/10.1093/toxres/tfae101>
- Chigor, V. N., Digwo, D. C., Adediji, A., Chidebelu, P. E., Chigor, C. B., Ugwu, K. O., Ibangha, I.-A. I., Street, R., & Farkas, K. (2024). Epidemiology of norovirus infection in Nigeria: A systematic review and meta-analysis. *Archives of Virology*, 169(7), 138. <https://doi.org/10.1007/s00705-024-06056-w>
- Daramola, A. S., & Akindureni, Y. (2017). Water quality analysis of Akure metropolis, Ondo State, Nigeria. *American Journal of Engineering Research*, 6(3), 28–31. [https://www.ajer.org/papers/v6\(03\)/E06032831.pdf](https://www.ajer.org/papers/v6(03)/E06032831.pdf)
- Egbueri, J. C., Ezugwu, C. K., Ameh, P. D., Unigwe, C. O., & Ayejoto, D. A. (2020). Appraising drinking water quality in Ikem rural area (Nigeria) based on chemometrics and multiple indexical methods. *Environmental Monitoring and Assessment*, 192(5), 308. <https://doi.org/10.1007/s10661-020-08277-3>
- Ekundayo, T. C., Igere, B. E., Oluwafemi, Y. D., Iwu, C. D., & Olaniyi, O. O. (2021). Human norovirus contamination in water sources: A systematic review and meta-analysis. *Environmental Pollution*, 291, 118164. <https://doi.org/10.1016/j.envpol.2021.118164>
- Ganiyu, S. A., Mabunmi, A. A., Olurin, O. T., Adeyemi, A. A., Jegede, O. A., & Okeh, A. (2021). Assessment of microbial and heavy metal contamination in shallow hand-dug wells bordering Ona River, Southwest Nigeria. *Environmental Monitoring and Assessment*, 193(3), 126. <https://doi.org/10.1007/s10661-021-08910-9>
- Gbadebo, A. M. (2021). Assessment of quality and health risk of peri-urban groundwater supply from selected areas of Abeokuta, Ogun State, Southwestern Nigeria. *Environmental Geochemistry and Health*, 43(7), 2743–2755.
- Ighalo, J. O., & Adeniyi, A. G. (2020). A comprehensive review of water quality monitoring and assessment in Nigeria. *Chemosphere*, 260, 127569. <https://doi.org/10.1016/j.chemosphere.2020.127569>
- Koko, D. T., Alfred, M. O., Bolujoko, N. B., Olorunnisola, D., Otitoju, O. B., Alabi, P., Ogunlaja, O. O., Okonofua, F., Omonkhua, A. A., Msagati, T. A. M., Omorogie, M. O., Ogunlaja, A., Olukanni, O. D., & Unuabonah, E. I. (2024). Prevalence and health risk evaluations of mycotoxins in drinking water sources in Nigeria. *RSC Advances*, 14(46), 34435–34447.
- Kraay, A. N. M., Man, O., Levy, M. C., Levy, K., Ionides, E., & Eisenberg, J. N. S. (2020). Understanding the impact of rainfall on diarrhea: Testing the concentration-dilution hypothesis using a systematic review and meta-analysis. *Environmental Health Perspectives*, 128(12), 126001. <https://doi.org/10.1289/EHP6181>
- Kumpel, E., Cock-Esteb, A., Duret, M., de Waal, D., & Khush, R. (2017). Seasonal variation in drinking and domestic water sources and quality in Port Harcourt, Nigeria. *The American Journal of Tropical Medicine and Hygiene*, 96(2), 437–445. <https://doi.org/10.4269/ajtmh.16-0175>
- Kusa, R., & Joshua, W. K. (2023). Evaluating the potability and human health risk of sachet water in Wukari, Nigeria. *Archives of Environmental and Occupational Health*, 78(2), 71–79. <https://doi.org/10.1080/19338244.2022.2063785>
- Nwokoro, U. U., Ugwa, O., Onwuliri, C. D., Obi, I. F., Murphy-Okpala, N., & Agunwa, C. (2020). Water, sanitation and hygiene risk factors associated with diarrhoea morbidity in a rural community of Enugu, South East Nigeria. *Pan African Medical Journal*, 37, 115.
- Oduori, D. O., Kwoba, E., Thomas, L., Grace, D., & Mutua, F. (2022). Assessment of foodborne disease hazards in beverages consumed in Nigeria: A systematic literature review. *Foodborne Pathogens and Disease*, 19(1), 1–18. <https://doi.org/10.1089/fpd.2021.0043>
- Olalemi, A. O., Ige, O. M., James, G. A., Obasoro, F. I., Okoko, F. O., & Ogunleye, C. O. (2021). Detection of enteric bacteria in two groundwater sources and associated microbial health risks. *Journal of Water and Health*, 19(2), 322–335. <https://doi.org/10.2166/wh.2021.212>
- Omeke, M. E., Ezugwu, A. L., Agbasi, J. C., Egbueri, J. C., Abugu, H. O., Aralu, C. C., & Ucheana, I. A. (2024). A review of the status, challenges, trends, and prospects of groundwater quality assessment in Nigeria: An evidence-based meta-analysis approach. *Environmental Science and Pollution Research*, 31(15), 22284–22307. <https://doi.org/10.1007/s11356-024-32552-2>
- Oyewale, A. T., Odetoyin, B. W., Oluduro, A. O., & Adeniyi, I. F. (2024). Occurrence of coliforms and biofilm-forming bacteria in raw, treated, and distributed water from two waterwork systems in Osun State, Southwestern Nigeria. *Journal of Water and Health*, 22(4), 673–688. <https://doi.org/10.2166/wh.2024.302>
- Rabiu, A. G., Falodun, O. I., Fagade, O. E., Dada, R. A., & Okeke, I. N. (2022). Potentially pathogenic *Escherichia coli* from household water in peri-urban Ibadan, Nigeria. *Journal of Water and Health*, 20(7), 1137–1149.
- Udoh, A., Lawal, B. K., Akpan, M., Labaran, K. S., Ndem, E., Ohabunwa, U., Tikare, O., Ibrahim, U. I., Amorha, K., & Kpokiri, E. (2021). Microbial contamination of packaged drinking water in Nigeria. *Tropical Medicine & International Health*, 26(11), 1378–1400. <https://doi.org/10.1111/tmi.13672>
- Uzuegbu, J. O., Ozoh, P. T. E., & Okuguni, C. C. (1996). Microbiological quality of drinking water in Akure metropolis, Nigeria. *International Journal of Environmental Health Research*, 6(3), 251–257. <https://doi.org/10.1080/09603129609356896>
- Wolf, J., Hubbard, S., Brauer, M., Ambelu, A., Arnold, B. F., Bain, R., & Boisson, S. (2022). Effectiveness of interventions to improve drinking water, sanitation, and handwashing with soap on risk of diarrhoeal disease in children in low-income and middle-income settings: A systematic review and meta-analysis. *The Lancet*, 400(10345), 48–59.
- World Health Organization, & United Nations Children's Fund. (2023). Progress on household drinking water, sanitation and hygiene 2000–2022: Special focus on gender. <https://www.who.int/publications/i/item/9789240076921>
- World Health Organization. (2022). Guidelines for drinking-water quality: Fourth edition incorporating the first and second addenda. World Health Organization.
- World Health Organization. (2024). Drinking-water. World Health Organization.
- World Health Organization. (2026). Guidelines for drinking-water quality: Small water supplies. World Health Organization.

Appendices

The figure illustrates variation in pH among the sampled water sources. The ODSWC_Ukere observations had the lowest mean pH, whereas the hand-dug well showed the highest mean pH (Fig. A1). The variation indicates differences in the chemical characteristics of the water sources.

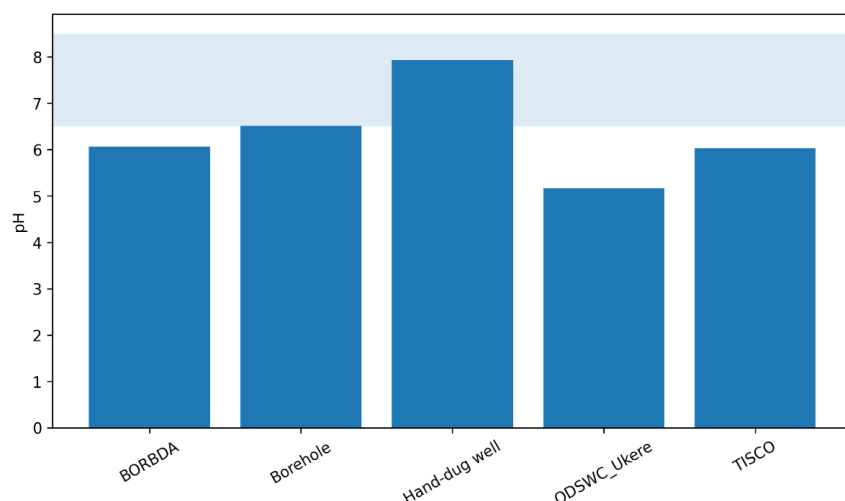


Figure A1. pH Distribution across Water Sources

EC and TDS varied considerably between the water sources. The highest EC was observed in the hand-dug-well group, while ODSWC_Ukere had the highest reported TDS (Fig. A2). These parameters provide an indication of the concentration of dissolved ionic and mineral substances in the water.

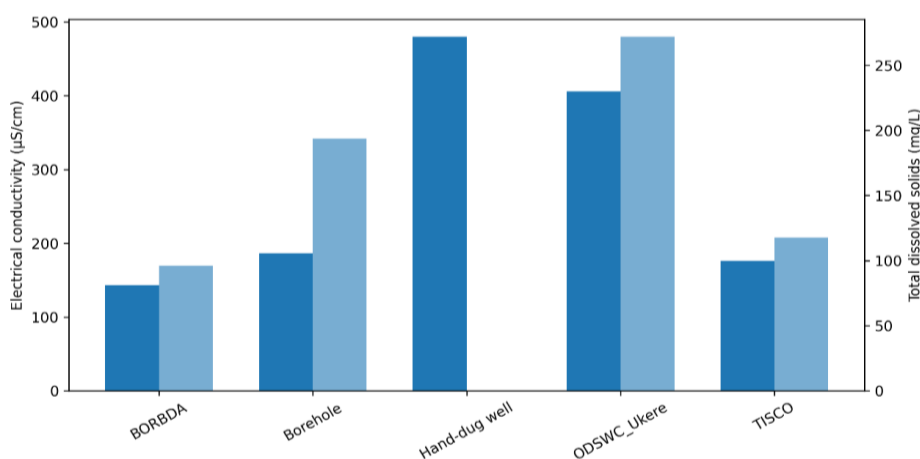


Figure A2. Electrical Conductivity and TDS across Water Sources

Turbidity varied across the water sources, with the borehole observations showing the highest mean turbidity among the groups represented (Fig. A3). Higher turbidity may reflect suspended particulate material in water. The figure therefore provides a visual representation of differences in the physical clarity of the sampled water.

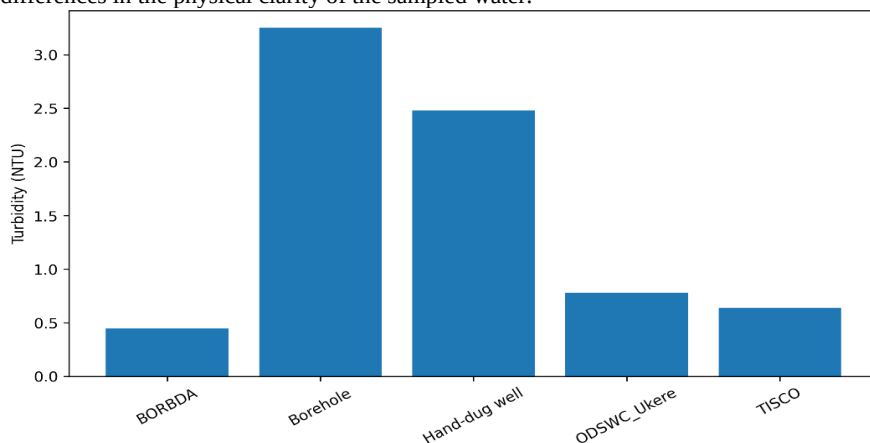


Figure A3. Turbidity Distribution across Water Sources

The microbiological indicators showed substantial differences among water sources (Fig. A4). BORBDA recorded the highest values for *E. coli*, total coliforms and faecal streptococci in the dataset. The presence of these indicators is relevant to the microbiological quality and sanitary condition of drinking-water sources.

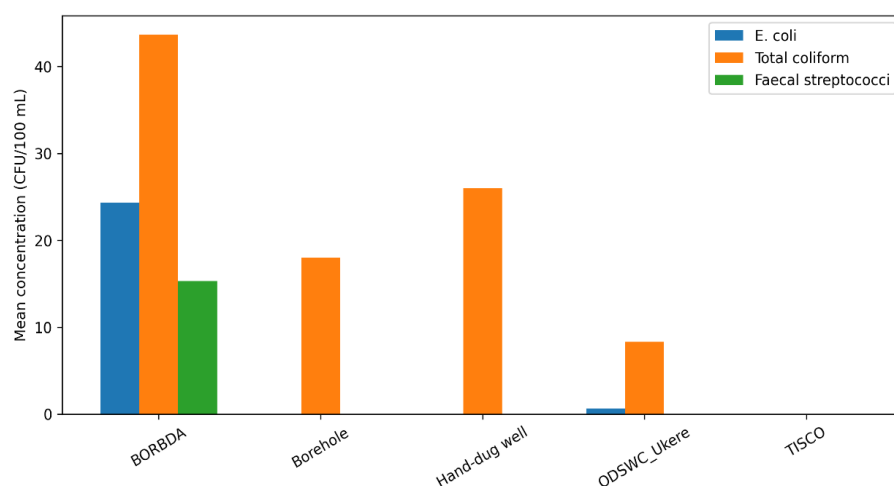


Figure A4. Microbiological Indicators across Water Sources

Rainfall varied considerably between the available annual records, while mean temperature remained comparatively stable (Fig. A5). The highest annual rainfall was recorded in 2021, whereas the available 2023 period had a lower cumulative value. The figure provides the climatic background for interpreting the environmental component of the study.

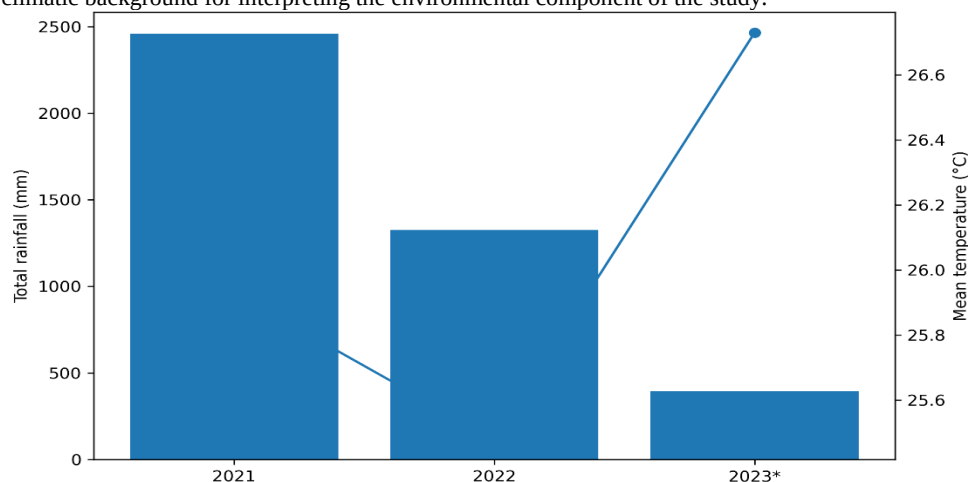


Figure A5. Annual Rainfall and Mean Temperature

Reported disease counts differed between Akure North and Akure South during June and July 2023 (Fig. A6). Typhoid reports were particularly high in Akure South in June and Akure North in July. The figure presents the surveillance pattern descriptively and does not imply a causal relationship with climate or water quality.

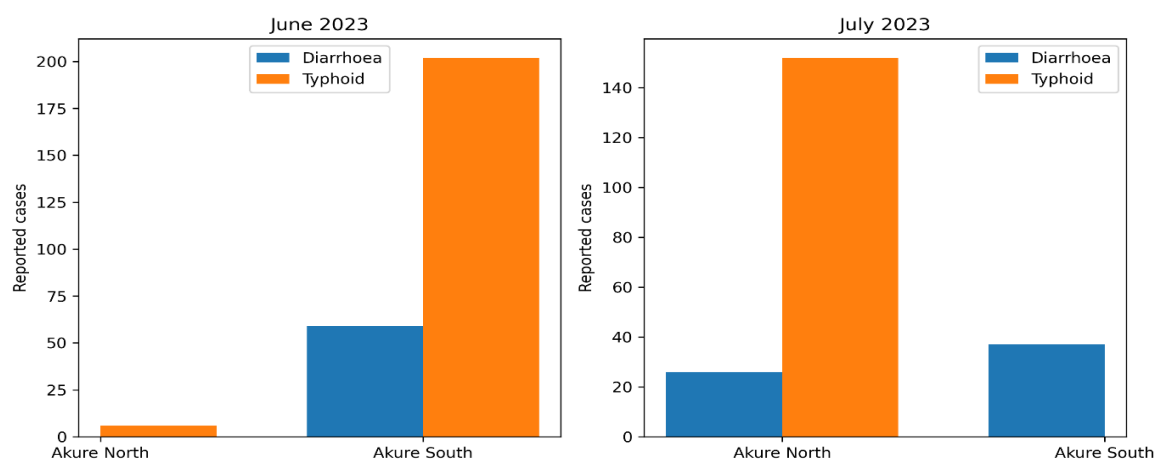


Figure A6. Disease-Surveillance Pattern by LGA

Figure A7 demonstrates that pH had the largest proportion of observations outside the specified operational criterion, followed by *E. coli*. Nitrate and turbidity had smaller proportions outside their respective criteria, while nitrite had none. These results provide a concise visual summary of the guideline-assessment findings.

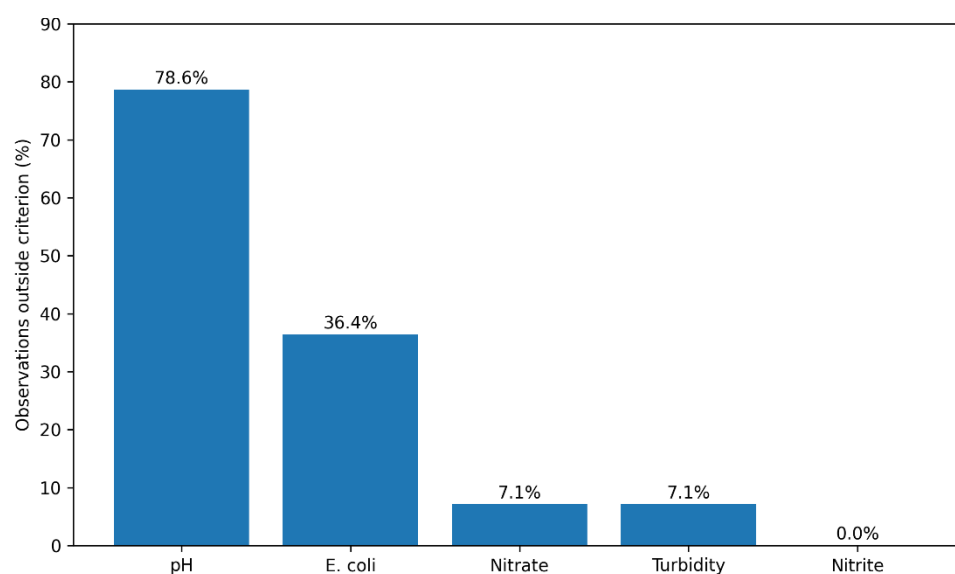


Figure A7. Water-Quality Observations outside Reference Criteria